

7.6 Factoring by Grouping

Lesson Introduction

 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Target	I can factor a polynomial by grouping.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.	
 Essential Question	How can you determine if factoring by grouping is the best method for factoring a polynomial?			
 Lesson Narrative	In this lesson, students explore the process of factoring by grouping. Students will learn how this method can be used when the coefficient of a term has multiple factors and when a polynomial has more than three terms. They will have opportunities throughout the lesson to practice factoring by grouping with a partner and independently, including working collaboratively to consider the process of factoring given polynomials.			
 Component Summary	7.6.1 Warm-Up (~5 min.)	Notice and Wonder activity with Coral Castle (<i>SE p. 387</i>)	7.6.4 Try It (10 min.)	Collaborative practice with factoring by grouping (<i>SE p. 389</i>)
	7.6.2 Guided Instruction (10-15 min.)	Formal instruction on factoring by grouping (<i>SE p. 387</i>)	7.6.5 Your Turn! (5 min.)	Independent practice with factoring by grouping (<i>SE p. 390</i>)
	7.6.3 Guided Practice (10 min.)	Guided practice with factoring by grouping (<i>SE p. 389</i>)	7.6.6 Wrap-Up (5-10 min.)	Ticket Out the Door with factoring (<i>SE p. 390</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A		N/A	
	<u>Additional Terms:</u> N/A			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recognize when a trinomial can't be factored and is prime. - Students may try to factor by grouping when it is not possible.	If time is an issue, consider assigning 7.6.5 Your Turn! for homework to allow adequate time for students to complete the 7.6.6 Wrap-Up for use in planning differentiated support before the end of the unit.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 7.6.1 Warm-Up provides students with an opportunity to determine what they notice and what they wonder about the sculptures at Coral Castle in Miami, Florida.	MA/K12.MTR.3.1: Mathematical Fluency 7.6.3 Guided Practice and 7.6.5 Your Turn! have students factoring polynomials by grouping. During these, students will select efficient and appropriate methods for solving problems. Additionally, they will complete these tasks accurately and with confidence. The lesson offers multiple opportunities for students to practice efficient and generalizable methods.
 Differentiate Instruction	Consider creating an anchor chart for student reference that highlights the process explored in 7.6.4 Try It with the provided example and then again with the example from question 2 from activity 7.6.3 Guided Practice.	
Lesson Notes:		

7.6.1 Warm-Up: Notice and Wonder

Component Overview: Students will review an image from Coral Castle in Miami, FL, to answer some questions about what they notice and wonder.



7.6.1 Warm-Up: Notice and Wonder

1. Coral Castle in Miami, Florida, was created by Edward Leedskalnin. He carved over 1100 tons of coral rock using an unknown process, often carving at night.



- a. What do you notice about the sculptures?

Answers will vary. I notice that there are multiple different shapes. I notice that some are surrounded by fencing and some are not.

- b. What do you wonder about the process Ed used?

Answers will vary. I wonder how he decided what shapes he would sculpt. I wonder how he knew what he could make out of that amount of coral rock.



After students have had an opportunity to view the image and answer the questions on their own, ask several students to share the things they noticed and things they wondered.

7.6.2 Guided Instruction: Factoring by Grouping

Component Overview: Students will delve deeper into factoring by grouping. Instruction includes factoring polynomials with more than three terms using this method.



7.6.2 Guided Instruction: Factoring by Grouping

When factoring $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c .

1. In both $8x^2 - 19x - 15$ and $8x^2 - 2x - 15$, the product of a and c is -120 .
 - a. List all the pairs of factors for -120 .
 $-1, 120; -2, 60; -3, 40; -4, 30; -5, 24; -6, 20; -8, 15; -10, 12;$
 $1, -120; 2, -60; 3, -40; 4, -30; 5, -24; 6, -20; 8, -15; 10, -12;$
 - b. Circle the factor pair for -120 that adds to the sum of -19 .
 $5, -24$
 - c. Draw a rectangle around the factor pair for -120 that adds to the sum of -2 .
 $10, -12$
2. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial.

$8x^2 - 19x - 15$	$8x^2 - 2x - 15$
$8x^2 - 24x + 5x - 15$	$8x^2 - 12x + 10x - 15$
$8x(x - 3) + 5(x - 3)$	$4x(2x - 3) + 5(2x - 3)$
$(8x + 5)(x - 3)$	$(4x + 5)(2x - 3)$

Factoring by grouping is helpful when factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors.



Students will have opportunities to practice in the remaining components of this lesson. The purpose of this component is to provide explicit instruction in factoring by grouping to ensure understanding. Guide students through the questions in this component as they follow along. Consider thinking aloud through each process while using talk moves like “Revoice” to have students make sense of the information in their own words as they formulate working definitions of factoring by grouping.



Consider turning question 1 into an interactive partner activity. Have students complete each part with a partner and then chart up all the pairs of factors for -120 onto a central location for all students to see and use in question 2.



The first two lines of the table can be confusing. The purpose of the first line is to rewrite the expression using the factor pair. The second line highlights that the commutative property can be used to rewrite the expression.

7.6.2 Guided Instruction: Factoring by Grouping (continued)

3. Factor $6x^2 - 19x + 15$ by grouping.

$$\begin{aligned} &6x^2 - 10x - 9x + 15 \\ &2x(3x - 5) - 3(3x - 5) \\ &(3x - 5)(2x - 3) \end{aligned}$$

Factoring by grouping can also be helpful when factoring a polynomial with four terms.

4. Factor $3y^4 - 6y^3 + 9y^2 - 18y$ by grouping.

$$3y(y - 2)(y^2 + 3)$$

5. The polynomial $3y^4 + 9y^2 - 6y^3 - 18y$ is a rearranged form of $3y^4 - 6y^3 + 9y^2 - 18y$.

Factor $3y^4 + 9y^2 - 6y^3 - 18y$ by grouping.

$$3y(y - 2)(y^2 + 3)$$

6. What do you notice about the factored forms of $3y^4 - 6y^3 + 9y^2 - 18y$ and $3y^4 + 9y^2 - 6y^3 - 18y$?

They are the same.



Consider posing one or more of the following discussion questions before wrapping up this component.

- How can factoring by grouping be useful when the leading coefficient has multiple factors?
- How can factoring by grouping be useful when a polynomial has four terms?
- Does writing the polynomials in standard form make a difference when factoring by grouping?

7.6.3 Guided Practice: Factoring by Grouping

Component Overview: Students will practice factoring by grouping in a structured environment.



7.6.3 Guided Practice: Factoring by Grouping

1. Factor $6x^2 + x - 35$ by grouping.

$$(3x - 7)(2x + 5)$$

2. Factor $-15g + 35gh - 7h + 3$ by grouping.

$$(7h - 3)(5g - 1)$$



Use informal observation from the previous component and lessons to inform how “guided” this Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.



When reviewing the answers as a whole group, consider prompting students to elaborate on their process and steps for each question. Position students to answer questions that ask more about the procedures and concepts rather than just “What did you get for your answer?”

7.6.4 Try It: Factoring by Grouping

Component Overview: Students will work with a partner to answer questions about the process of factoring by grouping. For this activity, students should work with a partner.



7.6.4 Collaboration: Factoring by Grouping

Work with your partner to complete the following.

- Factor $2x^2 + 3x + 4x + 12$ by grouping.
Not factorable. Prime.
- What happens when you try to group the expression and find a common factor?
The factored binomials are not equivalent.
- If you rearrange the middle two terms, what happens when you try to factor by grouping again?
The same results as before.

- Complete the statement.

When factoring an expression by grouping, group terms

that have coefficients that

- ☐ are exactly the same.
- ☒ have common factors.
- ☐ are in numerical order.

- Work with your partner to create a strategy for the best way to group an expression.
Answers will vary.



Release students to work on this activity from start to finish on their own with a partner. Once completed, review each question as a whole class. Engage students in a rich discussion around questions 2-4 to help them better understand the process. Have students then share out the strategies they developed in question 5.



For a challenge, have students describe what would happen with this process if the conditions of the incorrect choices in question 4 existed.

7.6.5 Your Turn!

Component Overview: Students will work independently to practice factoring by grouping.



7.6.5 Your Turn!

1. Factor $12x^2 - 16x - 3$ by grouping.
 $(6x + 1)(2x - 3)$



Encourage students to use a variety of methods that fit their strengths as natural scaffolding for attempting to factor by grouping.

Students who struggle with factoring by grouping may be able to guess and check efficiently to find the factors. Prompt those students to find a partner who did factor by grouping and make connections between their representation of grouping and the factors they found.

7.6.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses their knowledge of factoring. This activity should be done independently to provide data on student understanding.



7.6.6 Wrap-Up: Ticket Out the Door

1. Factor each expression.

a. $h^2 - 12h + 27$
 $(h - 9)(h - 3)$

b. $36a^4b^7 + 42a^6b^3$
 $6a^4b^3(6b^4 + 7a^2)$

c. $12x^2 - 8x - 15$
 $(6x + 5)(2x - 3)$

d. $5x^2 + 37x + 14$
 $(5x + 2)(x + 7)$



This Ticket Out the Door should be used to help determine students who need support on each type of factoring. This can serve as a “checkpoint” for the content covered thus far in this unit.

If time is a concern, consider assigning the 7.6.5 Your Turn! for homework to ensure adequate time for students to complete this Wrap-Up.